P2 - ADAPTING STHD MODEL TO IDENTIFY SPATIAL COMMUNITIES

Venkata Srikar Kavirayuni

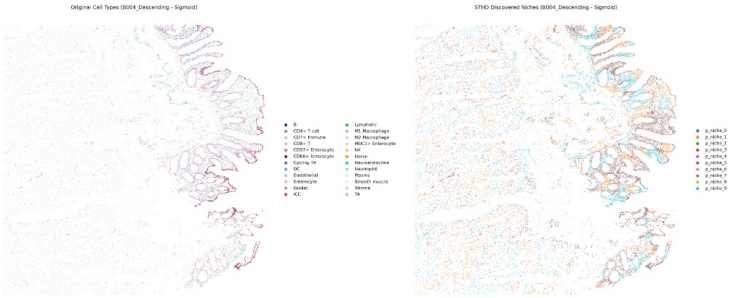
March 2026

1 P2 INTRODUCTION

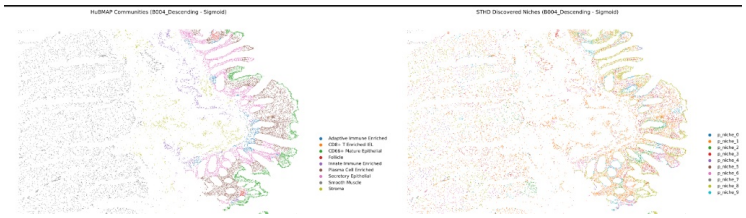
STHD is a statistical ML model which infers the maximum-likelihood estimate of cell type from ST data using a single-cell gene expression reference, assuming a Poisson distribution. Our model is an adaptation of that model which processes continuous spatial proteomics (CODEX) data alongside reference mean protein profiles to perform unsupervised discovery of multicellular spatial niches. We utilize a dual state STHD-modified architecture that simultaneously optimizes two latent variables: a spatial probability matrix assigning each cell to a specific niche and a composition matrix defining the fractional cell type makeup of those niches. The algorithm evaluates how well the expected protein signature of a predicted niche matches the observed continuous fluorescence of a cell to ground the spatial predictions in biological reality.

To handle continuous data our model replaces the original STHD Poisson count distribution with a variance scaled Gaussian log likelihood function. This biological likelihood is jointly optimized against a spatial cross entropy penalty governed by a physical Delaunay triangulation of the tissue. To prevent over smoothing across distinct anatomical structures we apply an anisotropic feature weighting to the graph edges using an exponential decay kernel based on protein space distance. This acts as a algorithmic constraint/rule making the model to enforce niche homophily within continuous tissue zones while preserving sharp heterophilic boundaries during optimization.

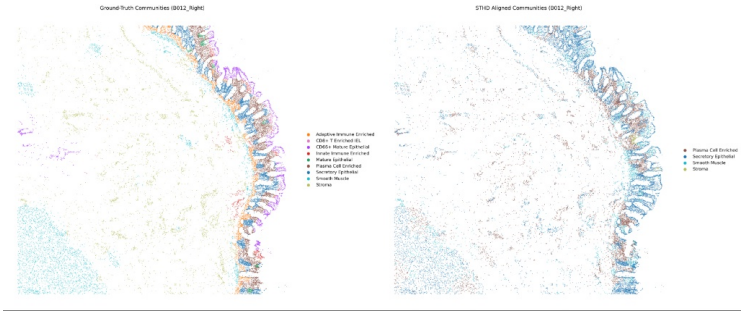
2 PRELIMINARY RESULTS



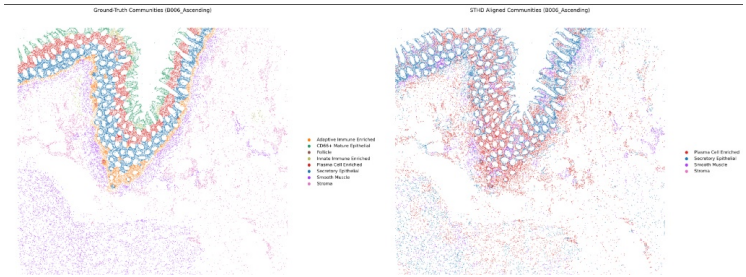
(a) B004 Descending Sigmoid, learning rate 0.1



(b) B004 Descending Sigmoid, learning rate 1.0



(c) B012 Right Colon, learning rate 1.0



(d) B006 Ascending Colon, learning rate 1.0

Figure 1: Empirical evaluation of STHD spatial niche discovery across three intestinal donors comparing the effects of composition matrix learning rates. Panel A shows donor B004 descending sigmoid processed with a restricted 0.1 rate. This restriction prevents the composition matrix from polarizing, causing niches to visually bleed across structural zones. Panels B, C, and D show donors B004, B012 right colon, and B006 ascending colon processed with an uncapped 1.0 rate. While the anisotropic edge weighting successfully preserves sharp epithelial boundaries, the high learning rate causes the biological likelihood to entirely dominate the spatial penalty. The model overfits by predicting pure single cell types instead of true multicellular blends. Compared to the ground truth HuBMAP communities, this overfitting fractures broad tissue zones into noisy disjointed segments, demonstrating the strict need for explicit entropy regularization.

Visual inspection of the mapped spatial niches shows that anisotropic edge weighting preserved the physical topology of the intestinal crypts. The structural firewalls effectively prevented the spatial penalty from smoothing across epithelial linings, keeping tissue borders intact. However, despite these preserved boundaries, community assignments are incorrect. The spatial maps appear fragmented, and STHD predictions fail to reflect the continuous tissue zones present in the HuBMAP ground truth.

The visual inconsistencies are supported by quantitative overlap metrics. The Adjusted Rand Index (ARI), which measures alignment between two clustering schemes (1.0 = perfect agreement, 0.0 = random overlap), remained near zero across all tested learning rates. At a learning rate of 1.0, the ARI was 0.01; at 0.5, 0.0099; and at 0.1, approximately 0.00. These results confirm that the current model architecture fails to identify correct multicellular combinations, independent of spatial regularization.

The failure arises from an imbalance between the biological log likelihood and the spatial cross entropy. The steep gradients in protein expression features from multiplexed CODEX data drive the optimization toward pure single-cell predictions. Minimizing the loss is achieved faster by selecting a single dominant cell type rather than maintaining balanced multicellular mixtures. As a result, the model overfits to local biological signal and ignores spatial relationships, reverting to standard single-cell typing rather than multicellular niche discovery.

This overfitting explains the mismatched colors observed in the visual outputs. The model produces clusters dominated by a single cell type, such as pure enterocytes or stromal cells. During visualization, each cluster is assigned a HuBMAP community label via majority voting. Because HuBMAP communities represent mixed cell-type regions, this mapping step produces visually inconsistent mosaics. The model therefore predicts biologically implausible structures with misplaced confidence.

so, ultimately in our current state, current results show that spatial cross entropy alone does not sufficiently enforce the learning of multicellular microenvironments. The optimization landscape favors pure biological fits over spatial coherence. To address this, additional regularization is required. Introducing an entropy penalty during composition matrix optimization will penalize pure cell-type outputs and encourage learning of mixed-cellular compositions consistent with HuBMAP ground truth.

What you have done already and are you on track I am currently on track with the project timeline. So far I have successfully adapted the original STHD algorithm to process continuous CODEX spatial proteomics data instead of discrete RNA counts. I replaced the original math with a Gaussian continuous likelihood function and implemented a Delaunay triangulation graph to better represent actual physical tissue layout. I also added anisotropic feature weighting to the spatial penalty. This successfully stopped the algorithm from over smoothing across sharp epithelial boundaries. Finally I mapped the unsupervised model outputs to the HuBMAP ground truth community labels to evaluate performance using the Adjusted Rand Index.

What are your plans going forward My immediate plan going forward is to fix the severe over fitting observed in the current model. Right now the model relies too heavily on the protein biology and ignores the spatial graph resulting in niches that just recreate pure single cell types. To force the model to learn true multicellular communities I will write an entropy penalty into the composition matrix optimization step. This will mathematically punish the model if it assigns a niche to just one cell type. Once the model stabilizes I plan to run baseline comparisons against other spatial clustering algorithms like CellCharter and BANKSY to finalize the benchmarking for the final report.

3 MATH DISCUSSION

1. The Original STHD Framework The original STHD algorithm was designed for spatial transcriptomics, specifically to denoise cell type assignments by leveraging physical proximity. Let X be the number of cells, Y be the number of genes, and Z be the number of reference cell types. The model infers a latent assignment matrix $P \in \mathbb{R}^{X \times Z}$, where $P_{a,t}$ is the probability that cell a belongs to cell type t.

The original biological likelihood assumed raw transcript counts $N_{a,g}$ followed a Poisson distribution parameterized by a reference mean profile $\mu_{t,g}$:

$$P(N_{a,g}|\mu_{t,g}) = \frac{\mu_{t,g}^{N_{a,g}} e^{-\mu_{t,g}}}{N_{a,g}!}$$

The log-likelihood objective maximized across the tissue was:

$$LL_{orig} = \frac{1}{X} \sum_{a=1}^{X} \sum_{t=1}^{Z} P_{a,t} \sum_{g=1}^{Y} (N_{a,g} \log \mu_{t,g} - \mu_{t,g})$$

To enforce spatial coherence, original STHD utilized a strict isotropic Cross-Entropy (CE) penalty based on a symmetric k-nearest neighbor graph. If $N(a)$ represents the neighbors of cell a, the spatial penalty enforces cell-level *homophily* (the assumption that neighboring cells should be the exact same biological type):

$$CE_{orig} = -\frac{1}{X} \sum_{a=1}^{X} \sum_{t=1}^{Z} P_{a,t} \sum_{a^* \in N(a)} \frac{1}{|N(a)|} \log P_{a^*,t}$$

The total objective optimized via Adam was:

$$Loss_{orig} = -LL_{orig} + \beta \cdot CE_{orig}$$

2. Dual-State Niche Discovery and Continuous CODEX Data To adapt STHD for highly multiplexed CODEX proteomics, two fundamental shifts were required. First, normalized continuous fluorescence intensity cannot be modeled via a Poisson distribution. We substituted the Poisson PMF with a Gaussian continuous log-likelihood. To prevent highly variable markers from dominating the gradient, the squared error is scaled by the global variance of each marker σ_g^2:

$$LL_{gauss} = -\frac{1}{2}\sum_{a=1}^{X}\sum_{t=1}^{Z} P_{a,t}\sum_{g=1}^{Y}\frac{(N_{a,g}-\mu_{t,g})^2}{\sigma_g^2}$$

Second, the biological goal shifted from cell-type denoising to multicellular spatial niche discovery. We introduced a Dual-State hierarchy . Let K be the number of latent spatial domains (niches). The model now infers $P_{niche} \in \mathbb{R}^{X\times K}$ and a dynamic composition matrix $\Theta \in \mathbb{R}^{K\times Z}$, where $\Theta_{k,t}$ represents the fractional composition of cell type t in niche k. To ensure Θ remains a valid probability distribution, we reparameterize it via an unconstrained weight matrix $V \in \mathbb{R}^{K\times Z}$ using a softmax function:

$$\Theta_{k,t} = \frac{\exp(V_{k,t})}{\sum_{z=1}^{Z}\exp(V_{k,z})}$$

The expected protein signature for a cell a in niche k becomes a linear combination of the underlying reference profiles:

$$E_{k,g} = \sum_{t=1}^{Z}\Theta_{k,t}\mu_{t,g}$$

The integrated Dual-State log-likelihood is:

$$LL_{dual} = -\frac{1}{2X}\sum_{a=1}^{X}\sum_{k=1}^{K} P_{niche}(a,k)\sum_{t=1}^{Z}\Theta_{k,t}\sum_{g=1}^{Y}\frac{(N_{a,g}-\mu_{t,g})^2}{\sigma_g^2}$$

3. Anisotropic Feature-Weighting for Boundary Preservation A critical failure of the initial spatial graph was the eradication of 1D epithelial boundaries (e.g., intestinal crypt linings). Originally, the model relied on a strict k-nearest neighbors graph ($k = 6$) with uniform edge weights ($w_{a,a^*} = 1.0$). This isotropic formulation enforces rigid *cell-level homophily* which is the mathematical assumption that a cell should be identical to the majority of its immediate spatial neighbors. In a 1D ribbon like the epithelium, a boundary cell touches only two other epithelial cells along the ribbon but up to four stromal cells beneath it. The uniform spatial cross-entropy (CE) penalty overwhelmingly favors the majority, mathematically ripping the boundary cell from its continuous ribbon and forcing it to merge into the massive 2D stromal blob.

Empirically, without anisotropic weighting, the continuous spatial cross-entropy penalty dominates the optimization landscape, causing catastrophic over-smoothing where the entire tissue collapses into just one or two dominant cell types, entirely erasing local biological heterogeneity.

To properly model tissue microenvironments, the algorithm must instead enforce *niche homophily* (neighbors should belong to the same broad spatial domain) while permitting *cell-level heterophily* (different cell types can live together within that domain, or be sharply separated by a structural boundary).

To achieve this, we first upgraded the topology from a rigid k-NN graph to a Delaunay triangulation, providing a biologically accurate, continuous physical mesh. Next, we modified the Delaunay edge weights w_{a,a^*} to be dynamically scaled by the protein-space distance between cells, creating an *anisotropic* penalty. Using the sum of variance-scaled squared differences

$$D_{a,a^*} = \sum_{g}\frac{(N_{a,g}-N_{a^*,g})^2}{\sigma_g^2},$$

we applied an exponential kernel tuned by the median distance $\tilde{D}$:

$$w_{a,a^*} = \exp\left(-\frac{D_{a,a^*}}{\tilde{D}}\right).$$

Under this exponential decay, if two adjacent cells share similar protein profiles, their distance D is near 0 and the edge weight $w \to 1.0$, fully enforcing spatial smoothing. Conversely, if an epithelial cell touches a stromal cell, their large protein distance drives the weight $w \to 0.0$ (effectively severing the connection).

This transforms the isotropic CE penalty into an anisotropic boundary-aware penalty:

$$CE_{aniso} = -\frac{1}{X}\sum_{a=1}^{X}\sum_{k=1}^{K} P_{niche}(a,k)\sum_{a^*\in N(a)} w_{a,a^*}\log P_{niche}(a^*,k).$$

This acts as a mathematical firewall. It allows the model to smooth probabilities along structurally continuous biological ribbons (where $w \approx 1.0$) while physically severing the spatial penalty across sharp, heterophilic structural boundaries (where $w \approx 0.0$).

4. The Failed 3-Headed Hierarchical Joint Model An attempt was made to jointly optimize explicit single-cell types P_{ct} and regional niches P_{niche} simultaneously. This required a three-headed optimization approach, introducing a linkage penalty (Kullback-Leibler divergence) to force agreement between the explicit cell type probability and the expected cell type distribution derived from the niche assignment . The expected cell type distribution for cell a based on its niche neighbors was defined as:

$$E_{ct}(a,t) = \sum_{k=1}^{K} P_{niche}(a,k)\Theta_{k,t}$$

The model optimized three simultaneous losses:

$$LL_{prot} = \frac{1}{X}\sum_{a=1}^{X}\sum_{t=1}^{Z} P_{ct}(a,t)F_{a,t}$$

$$CE_{space} = -\frac{1}{X}\sum_{a=1}^{X}\sum_{k=1}^{K} P_{niche}(a,k)\sum_{a^*} w_{a,a^*}\log P_{niche}(a^*,k)$$

$$CE_{link} = -\frac{1}{X}\sum_{a=1}^{X}\sum_{t=1}^{Z} P_{ct}(a,t)\log E_{ct}(a,t)$$

The objective function was $Loss_{joint} = -LL_{prot} + \beta CE_{space} + \gamma CE_{link}$.

This architecture ultimately proved highly unstable. The high-dimensional protein data provided steep, decisive gradients for LL_{prot}, causing P_{ct} to converge rapidly. Consequently, to minimize γCE_{link}, the model aggressively updated P_{niche} to strictly align with P_{ct}, entirely bypassing the spatial regularization matrix. The spatial penalty CE_{space} exploded as the model sacrificed all spatial smoothness to minimize the biological linkage disagreement.

5. Dual-State Reversion and Entropy Regularization Given the instability of the joint objective, we revert to the Dual-State (Niche-only) model defined in Section 2, which successfully preserved anisotropic boundaries. However, we observed that Θ collapses into near-binary vectors (e.g., $\Theta_{k,t} \approx 0.99$ for a single cell type), failing to model true multicellular communities. During empirical tuning of the Θ matrix (via the unconstrained weights V), we observed a strict trade-off dependent on the Adam optimizer's step size. When the learning rate for V was restricted (e.g., 0.1), the composition matrix failed to polarize, resulting in blurry, high-entropy mixtures that visually manifested as blocky, fragmented patches bleeding across boundaries. Conversely, when the learning rate was uncapped (e.g., 1.0), the biological likelihood overpowered the spatial graph, forcing $\Theta_{k,t}$ to rapidly collapse to near-binary vectors (≈ 0.99 for a single, often incorrect, cell type), failing to model true multicellular communities. Furthermore, we found that the model is highly sensitive to the total number of iterations. While 25 iterations preserved local community structure, extending the optimization to 125 iterations allowed the spatial penalty to gradually homogenize the graph, inevitably collapsing the entire tissue map back into a single cell type. This confirms that early stopping and explicit regularization are strictly required.

To explicitly prevent Θ from converging to pure cell types, we introduce an Entropy Penalty to the optimization of V. The Shannon entropy of the composition vector for niche k is:

$$H(\Theta_k) = -\sum_{t=1}^{Z}\Theta_{k,t}\log\Theta_{k,t}$$

To encourage heterophilic, multicellular mixtures, we must maximize this entropy (i.e., minimize the negative entropy). The regularized objective function becomes:

$$Loss_{final} = -LL_{dual} + \beta\cdot CE_{aniso} + \lambda\sum_{k=1}^{K}\sum_{t=1}^{Z}\Theta_{k,t}\log\Theta_{k,t}$$

Where λ controls the uniform-mixing prior. The gradient of this new penalty with respect to the unconstrained weights V will directly counter the model's tendency to sparsify the Θ matrix.

Alternatively, a Dirichlet prior could be placed over Θ to favor specific known multicellular compositions, or Temperature Scaling could be applied to the softmax function ($\Theta_{k,t} = \exp(V_{k,t}/\tau)/\sum\exp(V_{k,z}/\tau)$ with $\tau > 1$) to artificially soften the distributions without altering the loss landscape directly. We proceed with the explicit Entropy Penalty for granular gradient control.